# notes from Portfolio Selection - Efficient Diversification of Investments (Harry M. Markowitz)

Portfolio Selection: Efficient Diversification of Investments, 1959
https://www.jstor.org/stable/j.ctt1bh4c8h

## Section 1 - intro

A good portfolio is a balanced whole, providing the investor with protections and opportunities with respect to a wide range of contingencies.

The investor should build toward an integrated portfolio which best suits his needs. This monograph presents techniques of Portfolio Analysis is directed toward determining a most suitable portfolio for the large private or institutional investor.

A portfolio analysis starts with information concerning individual securities. It ends with conclusions concerning portfolios as a whole. The purpose of the analysis is to find portfolios which best meet the objectives of the investor

How do we determine whether to use that individual securities before deciding in portfolio?

- past performance
- thru opinions of security analysts (financial advisors) - predictions of future performance (formed via past performance/evaluation of existing/past portfolios)

**uncertainty**
even with predictions/forecast, there is always uncertainties (non-economic factors can influence perofmrance e.g. health of president or a success of a particular security - tariffs lol) - can affect capital gains / the market

you can't expect too much from security analysts (even if you provide them with too much information including confidential/insider information)

This security may be expected to do well if securities in general do well. It must be expected to do poorly if securities in general do poorly. Even this following of the market is not certain. There are weaknesses which may cause it to do poorly even though securities in general are performing well: The possibility of a labor dispute or of an aggressive competitor cannot be ignored. On the other hand, there are potentialities which may bring success greater than even the corporation management dares hope. The new styling of the product, the (not inexpensive) advertising campaign, and the expansion of production facilities may prove to be a magic combination, fulfilling all expectations for it.

however you can use uncertainty to make informed opinions about security performance (forecasted) to evaluate the strength/weakness of each individual security to analyse portfolios

The return (including capital gains and dividends) on security A is less uncertain than that on security B; the return on security C is more closely connected to the course of the general market than is that on security D; the growth of security E is more certain but has less potential than that of security F; only if the demand for their industry's product continues to expand (as it is likely, but not certain, to do) will the return on securities G and H be satisfactory.

**correlation among security returns**

- returns on securtity (stock movement) tend to move up/down together - it determines the market/economic acitivity
- there are some occasions where it may move against general flow (what is expected)
- we can observe other securities' performance - where they belong in SAME industry (correlation is NOT same for all industries)
- however if security returns are not correlated, diversification can eliminate risk (they can cancel out), but having CORRELATION will prevent potential highs/lows in investment to CANCEL out
- however if we want to reduce the risk, it is necessary to avoid a portfolio that is highly correlated - hence the need for diversification (its riskier to have all securities that perofmr the same - lets say if all of them go down, itll result in money being lost) . it is better for high/lows to cancel out than to have all lows
- TLDR: **diversification reduces risk**

**objectives of portfolio analysis**

criteria depends on the individual (what is the aim of the portfolio - what does the investor want to prioritise/how risk-adverse is the investor). there is no objective/standard way to evaluate portfolios (but it depends on the criteria set/nature of the investor)

common objectives

- high returns
- returns to be dependable, stable and not subject to uncertainty

however we can refer to portfolios by **efficient** or **inefficient**, by evaluating the likely returns / uncertianty and how it fits the investor's requierments (a portfolio with higher returns/lesser uncertainty/meets investor requirements is deemed more efficient than the other portfolio)

- where higher returns can have higher uncertainty (decision to invest depends on how the investor views risk)

**techniques of portfolio analysis:**

- separate efficient/inefficient portfolios (ignore the inefficient ones)
- portray combinations of likley return and uncertainty of returna vailable from efficient portfolios
- have the investor or investment manager carefully select the combination of likely returna nd uncertianty that best suit his needs
- determine portfolio which proviodes most suitable combination of risk and return (tradeoff)

**how do we select a desirable prortfolio**

- analyse each security from a long list of promising securities (view them as candidates to put in a desired portfolio)
  - can anaylse by calculating the return (closing price - last year closing price + dividends), divided by last year's closing price
  - thta will be the amount an investor would have made or lost if he invested $1 at the end of the last year and collected dividends declared in current year and sold at the end fo current year. profit = positive return, loss = negative return
  - e.g.

    **For example, if the closing price of 1947 were 50, that of 1948 were 45, and $2 of dividends were declared during 1948, then the return in 1948 would be**

    $$\frac{45 - 50 + 2}{50} = -.06,$$

    **or a loss of 6% per dollar invested.**
  - (portfolio selection should be ased on reasonable belifs about future performance, rather than relying on past perforamnces per se), however we still can use that as a basis (as we can estimate
    - the **likely return** based on average returns of past performance
    - the **uncertainty of return** in future by measuring variability of return in past

    - we can find the return for any combination of securities in any year (based on the distrbituion of securities/assets)\
    - instead of measuring greatest (maximum) loss for variability of return, we can use the **standard deviation**
        - difference between average return and observed return, squared and then averaged will obtain the **variance** from the average (similar to MSE but for deviations from average)
        - the average squared deviation is the **variance**; the **square root of average square deviation** is **standard deviation**
        - standard deviation is NOT only determined by std of individual securities, also depends on correlations between securities (correltaion coefficient measures how two sets of numbers tend to move up/down together - COEF 1: move up/down perfect unision, 0: uncorrelated (no correlation))
        - the moer two series of numbers tend to mve up or down together the greate rhte correlation coefficient
    - standard deviation of portfolio is determined by:
        - std of each security
        - correlation between each pair of securities
        - amount invested in each security
        - (higher corelation => greater the standard deviation)
        - tldr: more returns on individaul securities tend to move up or down together, the lesser the variations in inidivudal securities cancel out each other, hence greater variability of return
        - lesser the standard deviation, the less volatile a portfolio is (greater stability)
- plot a curve to compare portfolios (you compare average return / standard deviation)
    - on the curve - efficient
    - not on curve - inefficient (above the curve)
- based on curve, investor must decide on the most efficient combination on average/standard deviation which satisfies his needs/preferences with respect to riska nd return (how risk adverse he is)
- after investor decides, then we can start identifying all corner portfolios

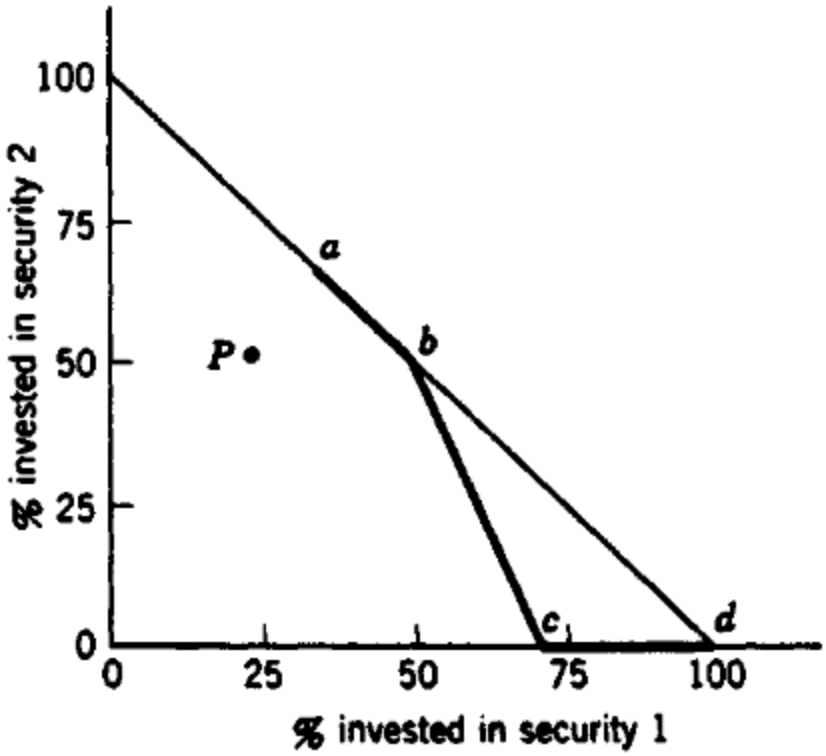


**Figure 5. Efficient portfolios among three securities.**

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- (corner portfolios), key turning points along the efficient frontier where the composition of the optimal portfolio changes significantly. They represent portfolios where:
  - At least one asset either enters or exits the portfolio (hits a constraint)
  - Or an asset reaches its maximum or minimum weight limit
- we can use corner portfolios to optimise/find weights to use that get us the best average return.stadnadrd deviation of return on the curve
- we could also use probability beliefs of experts as inputs to a portfolio analysis (security analyst - his statements of future of a security will be based on general conditions, prospects for economy and makret, nature of psosible new developments in industry, past performance, financial structures, position of security, opportuntiies/dangers confronting corprotation)
- The information required for the portfolio analysis depended on
  - (1) the objectives of the investor and
  - (2) the need of the portfolio analysis for estimates of the most likely return on each security, the uncertainty of return associated with each security, and the correlation between each pair of securities.
- a desirable portfolio
  - (i) would yield at least a specified current income,1 to provide for current expenses, and

    - (ii) would grow in value (from additional income or capital gains) during the next three years, to provide the basis of future income and growth.
- returns on most securities are correlated, where if S&P (Standard and Poor) index rises, we can expect other indexes to rise too (highly correlated).less tha normally correlated -> corretiaon does not fit reltaionship with the index - e.g. security moves opposite to other if index stayed contant). normally correlated -> returns of two securities are correlated only because they are correlated with things in general (and S&P in particular)

# Section 2 - relationships between securities and portfolios

## Averages and Expected Values

The statistics will cover these relationships to consider: (identical in form but different in substance)

1. relationships among past returns on securities and those on portfolios
2. relationships among probability beliefs of securities and their implications for portfolios
3. relationships among **random variables**
   **Random variables**: a number generated by a chance device e.g. the number of dots appearing after a roll of dice / no of aces drawn in hands of cards

- we can apply that concepts to probability beliefs

**Frequency Distributions**
Tells us how the frequency is distributed
Can derive **relative frequency** of each occurence

**Probability Distribution**
Tells us how probability is distributed among possible outcomes
Contains all pertinent information containing a random variable

TABLE 2

PROBABILITY BELIEFS

| Return | Probability |
|---|---|
| −.30 | 1/30 |
| −.20 | 2/30 |
| −.10 | 4/30 |
| .00 | 6/30 |
| .10 | 8/30 |
| .20 | 5/30 |
| .30 | 2/30 |
| .40 | 1/30 |
| .50 | 1/30 |

Tabular presentation of (hypothetical) probability beliefs.

**Average / Expected Value**

Average Value: Sum of series divided by number of items in series

Expected Value: The sum of the (value of outcome times its probability) for each outcome

TABLE 5

DERIVATION OF EXPECTED VALUE FROM PROBABILITIES

| (1) Outcome | (2) Probability | (3) (1) · (2) |
|---|---|---|
| −.10 | 1/8 | (−.10)/8 |
| −.05 | 1/8 | (−.05)/8 |
| .00 | 2/8 | .00 |
| .05 | 3/8 | .15/8 |
| .10 | 1/8 | .10/8 |
| SUM | 1.00 | .10/8 = .0125 |

Expected value is NOT the value to 'expect' (but is just the weighted average of possible outcomes with probabilities of occurrence used as weights)

we don't EXPECT that value as outcome

**Alternative Measures of Central Tendency**
Averages/Expected values measure **central tendency** (where the center of a distribution is located)

- average past return = number about which past returns fluctuated
- expected future return = centrally located with respect to future possible outcomes

otherw ways to measure:

TABLE 7

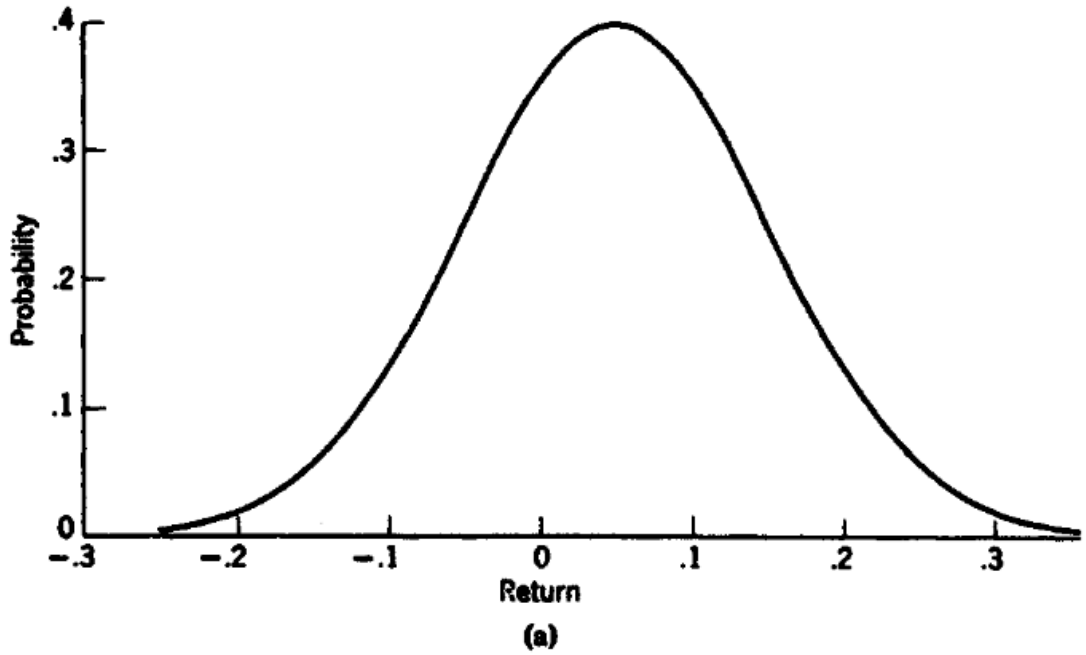
THREE MEASURES OF CENTRAL TENDENCY

| Technical Name | Alternative Name, past series | Alternative Name, random variables | Description |
|---|---|---|---|
| Mean | Average | Expected value | A weighted average using frequencies or probabilities as weights |
| Mode | Most frequent | Most likely | That number (or those numbers) which appears more frequently, or has a higher probability than any other |
| Median | Midpoint | Midpoint | A number such that 50% of the observations (or 50% of the probability) are associated with numbers equal to or greater than this number, and 50% are associated with numbers equal to or less than this number |

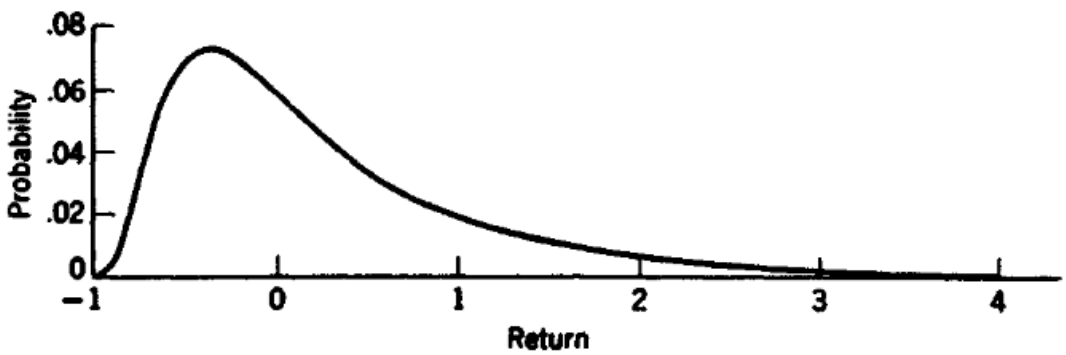
several different combinations of measures can elad to same set of efificent portfoolios, where it can be true if probabiltiy distribution of returns follow a normal distribution OR log-nomral distribution. this can mean that a portfolio is mean-std-efficient if and only if its

- mode - probability of loss efficient
- mode - std efficient
- mean - probability of loss efficeint
- median - std efficent and
- median - probability of loss efficient

in this case the choice of measures depends on conveneince



(a)



(b)

(normal distribution vs lognormal distribution)

a distribution can never have moer than 1 mean, but can have multiple median/medians

**Constant times Random Variable**
Suppose $r$ is any random variable generated by any wheel; suppose $A$ is any number
suppose $Ar$ is athe random variable which is always $A$ times as large as $r$
then the expected value of $Ar$ is $A$ times as great as the expected value of $r$

**Random Pairs**
Suppose $q$ and $r$ are a pair of random variables generated by any wheel. Suppose $s$ is the random variable which always equal to the sum of $q$ and $r$. Then the expected value of $s$ equals to the expected value of $q$ plus the expected value of $r$

(This theorem is asserted for any pari of random variables - whether they are correlaetd or uncorrelaetd)

**Expected Value of Weighted Sum of Two Random Variables**
Suppose that $q$ and $r$ are any pair of random variables generated by any wheel; suppose that $A$ and $B$ are any numbers; suppose, finally that $w$ is the random variable which is always equal to $Aq + Br$.

The expected value of $w$ equals $A$ times expected value of $q$ plus $B$ times expected value of $r$

Example:
If security I has had an average return of 12% and security II has had an average return of 8%, then the average return per dollar invested in a portfolio consisting of 1 /4 security 1 and 3/4 security II was $(1/4 * 12)$ + $(3/4 * 8)$ - 9%.

**Weighted Average of any Number of Random Variables**
Suppose that $r_1, r_2, r_3, \ldots, r_N$, is any set of $N$ random variables generated by some wheel. Suppose that $A_1, A_2, A_3, \ldots, A_N$ is any set of $N$ numbers. Suppose that $w$ is the random variable which is inevitably equal to $A_1 r_1 + A_2 r_2 + A_3 r_3 + \ldots + A_N r_N$. Then the expected value of $w$ is

- ($A_1$ times the expected value of $r_1$ plus
- ($A_2$ times the expected value of $r_2$ plus
- ($A_3$ times the expected value of $r_3$ plus
- (and so on, until)
- ($A_N$ times the expected value of $r_N$).

**Results**

- Expected value of a weighted sum is the weighted sum of expected values.
- True whether random variables are correlaetd or uncorerlaetd
- True for any number of random variables, no mtter how they move up or down together

The theorem of the last section can be rephrased to apply to past series or to uncertain future events:

1. If a series ($S$) is the weighted sum of $N$ other series, then the average of series $S$ is the same weighted sum of the averages of the series.
2. If an individual believes that the value ($V$) of some future event must inevitably equal a particular weighted sum of the values of $N$ other future events, then, if his probability beliefs are internally consistent, the expected value of $V$ must be the weighted average of the expected values of the $N$ events.

# Standard Deviations and Variances

**Variance**

- Average of squared deviations (distance between average return and actual return)

**Standard Deviation**

- The square root of the variacne
- Measure of variability that takes more into account than just one or two exterme returns

You can use standard deviation to compare the return's distance from average (e.g. approx 2 standard deviations below average)

Variance of random variable

- Expected value of squared deivation from expected value
- Can be computed from table of probabilities

Standard Deviation of a random variable measures how close the random variable is likely to be to its expected value (its the square root of its variance)

In probability beliefs, standard deviation can be used to measure the degree of uncertianty associated with future event

**Constant Times a Random Variable**

# Investment in Large Number of Securities